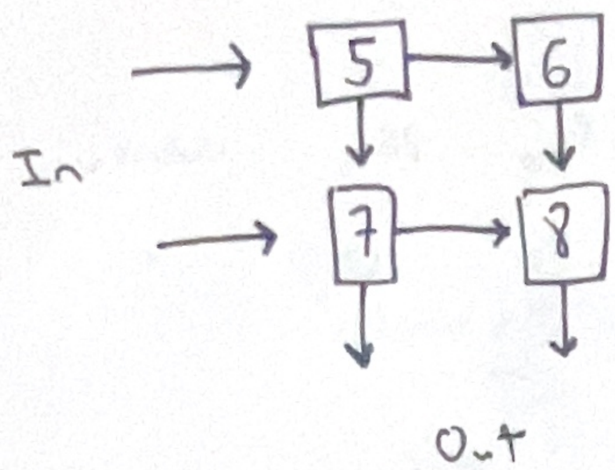
ECE 510 S26 Naveed Scharmszadeh CMAN 5

1) $AB = C$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$



Feed columns of A into systolic array.

On cycle 1, broadcast A_{00} to PE_{00} and PE_{01}.

~~Cycle~~ Cycle 2: broadcast A_{10} to PE_{00} and PE_{01}

broadcast A_{01} to PE_{10} and PE_{11}

2)

Cycle	Input row 0	Input row 1	PE_{00}	PE_{01}	PE_{10}	PE_{11}	Output C
1	1	0	1·5	1·6	0	0	x
2	3	2	3·5	3·6	1·5+2·7	1·6+2·8	x
3	0	4	0	0	3·5+4·7	3·6+4·8	$\begin{pmatrix} 19 & 22 \\ 0 & 0 \end{pmatrix}$
4	0	0	0	0	0	0	$\begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$

<u>Note:</u> From consulting Claudian I mentioned how the rows of C may be interchanged based on how the systolic array stores the outputs. It told me that the order of the rows does not really matter as long as I get the correct computed numbers.

3) a) Total MACs (not including zero-padding): 8 (over 4 cycles)

b) Each input is reused once.

c) Off-chip memory accesses: 1 for A, 1 for B, 1 (write) for C.

4) For output stationary dataflow, the final output elements of C stay fixed in the PEs.